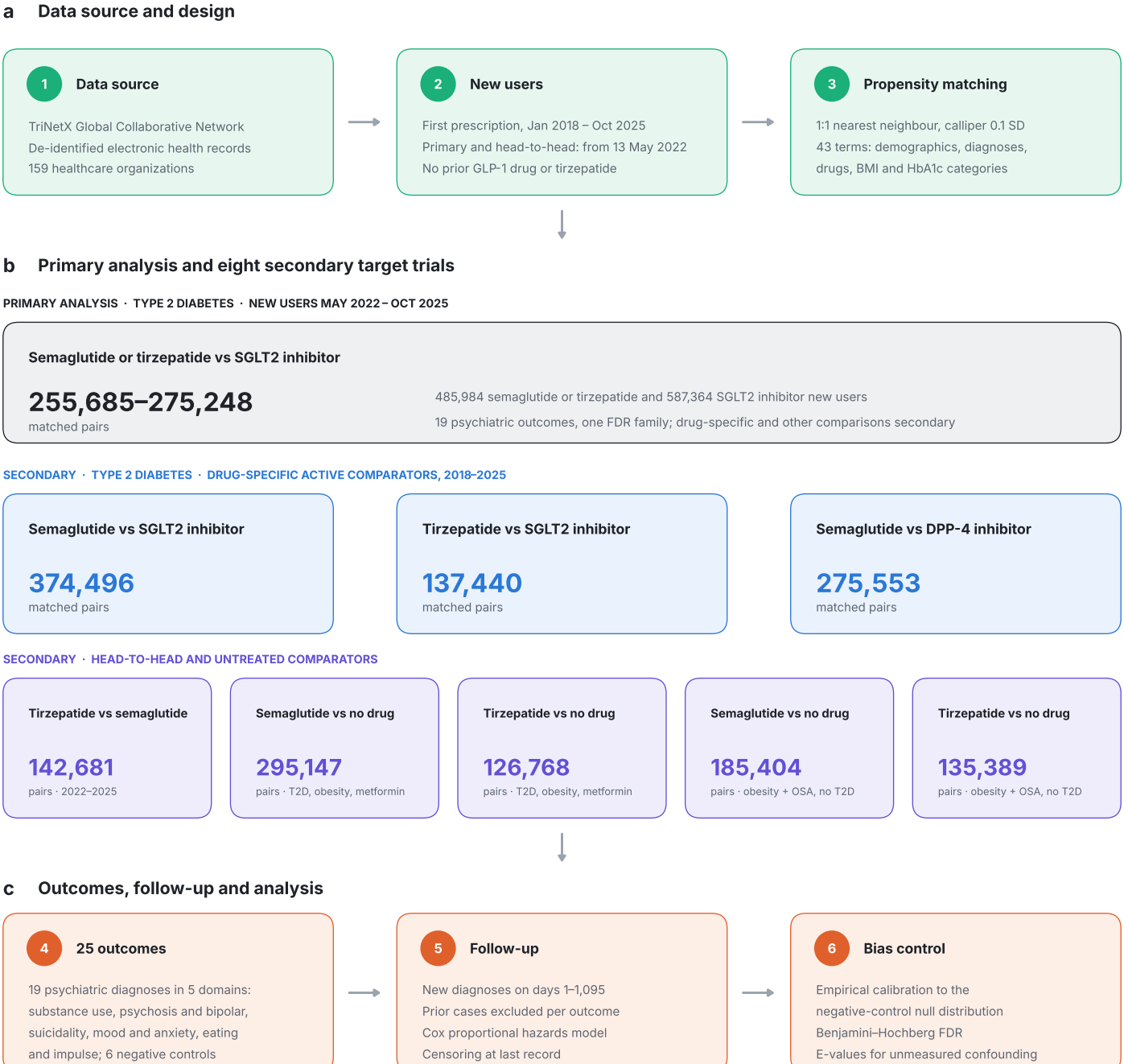
a
Data source and design
1
Data source
TriNetX Global Collaborative Network
De-identified electronic health records
159 healthcare organizations
2
New users
First prescription, Jan 2018 – Oct 2025
Primary and head-to-head: from 13 May 2022
No prior GLP-1 drug or tirzepatide
3
Propensity matching
1:1 nearest neighbour, calliper 0.1 SD
43 terms: demographics, diagnoses,
drugs, BMI and HbA1c categories
b
Primary analysis and eight secondary target trials
PRIMARY ANALYSIS · TYPE 2 DIABETES · NEW USERS MAY 2022 – OCT 2025
Semaglutide or tirzepatide vs SGLT2 inhibitor
255,685–275,248
matched pairs
485,984 semaglutide or tirzepatide and 587,364 SGLT2 inhibitor new users
19 psychiatric outcomes, one FDR family; drug-specific and other comparisons secondary
SECONDARY · TYPE 2 DIABETES · DRUG-SPECIFIC ACTIVE COMPARATORS, 2018–2025
Semaglutide vs SGLT2 inhibitor
374,496
matched pairs
Tirzepatide vs SGLT2 inhibitor
137,440
matched pairs
Semaglutide vs DPP-4 inhibitor
275,553
matched pairs
SECONDARY · HEAD-TO-HEAD AND UNTREATED COMPARATORS
Tirzepatide vs semaglutide
142,681
pairs · 2022–2025
Semaglutide vs no drug
295,147
pairs · T2D, obesity, metformin
Tirzepatide vs no drug
126,768
pairs · T2D, obesity, metformin
Semaglutide vs no drug
185,404
pairs · obesity + OSA, no T2D
Tirzepatide vs no drug
135,389
pairs · obesity + OSA, no T2D
c
Outcomes, follow-up and analysis
4
25 outcomes
19 psychiatric diagnoses in 5 domains:
substance use, psychosis and bipolar,
suicidality, mood and anxiety, eating
and impulse; 6 negative controls
5
Follow-up
New diagnoses on days 1–1,095
Prior cases excluded per outcome
Cox proportional hazards model
Censoring at last record
6
Bias control
Empirical calibration to the
negative-control null distribution
Benjamini–Hochberg FDR
E-values for unmeasured confounding